

AI + DSA Roadmap for B.Tech (3rd Year)

Goal: Build strong placement-ready skills in AI/ML/GenAI while mastering DSA.

Priority 1: DSA

Arrays, Strings, Linked Lists, Stacks, Queues, Trees, Graphs, Dynamic Programming. Solve 2–3 problems daily.

Priority 2: AI/ML

Python, NumPy, Pandas, Matplotlib, Scikit-learn, PyTorch, Deep Learning.

Priority 3: GenAI

Prompt Engineering, LLMs, RAG, LangChain, AI Agents, Vector Databases.

Priority 4: Projects

Brain Tumor Detection, AI Content Detection, RAG PDF Chatbot, AI Interview Assistant, Multi-Agent Research Assistant.

Weekly Schedule

Mon–Fri: DSA (1–1.5 hr), AI/ML (2 hr), Project Work (1 hr).

Sat–Sun: Projects, GitHub, Resume, LinkedIn, Mock Interviews.

Targets

300+ DSA problems, 4 strong AI projects, GitHub portfolio, LinkedIn profile, 1–2 internships/certificates.